

**Comprehensive Exams: SVAR and Impulse Response Functions**

Bing Cai Kok

University of North Carolina at Chapel Hill

### Comprehensive Exams: SVAR and Impulse Response Functions

*Provide an overview of multiple ways that SVAR models can be used to calculate impulse response functions. Include in your answer methods currently used within Psychology as well as those you found in other fields. When providing your review, include critiques regarding the assumptions made and practicality of the methods for future psychological studies. End with a set of recommendations for researchers of psychological studies.*

### Introduction

Technological advances have radically changed the way psychologists collect their data. While early studies in psychology have been restricted to repeated measurements with only a few timepoints (e.g. panel data), modern psychological data can often involve repeated measurements that span hundreds (or even more) of timepoints. The popularity of such data is growing rapidly throughout psychology. For example, the use of functional Magnetic Resonance Imaging (fMRI) often entails the collection of several hundred repeated measurements of neural signals per person (Beltz & Gates, 2017). Similarly, clinical studies increasingly use Ecological Momentary Assessment (EMA) to sample people's emotions across multiple days and months (Wright et al., 2015). In either case, such time-series data enable psychologists to now probe within-person processes in much finer detail than ever before. The intensive longitudinal characteristic of such data, however, calls for proper methodology for the analysis of such data.

The main issue addressed in this writeup is on the analysis and interpretation of such time-series data. Our discussion is structured as follows. We will first discuss the standard Vector Autoregressive Model (VAR) used in much of psychological work. Thereafter, we will consider the problem of interpreting such VAR models through the lens of impulse responses. In particular, we will highlight how psychological questions that are framed through a *structural* interpretation of a VAR are plagued by an identification problem. We will then consider different ways through which this issue is tackled in both

psychology and econometrics. Finally, we will provide a set of recommendations for the use of impulse responses within the context of psychology.

### Vector Autoregressive Models

We are interested in data with the following specification. Let  $K$  be the number of variables and  $T$  be the number of timepoints. The time-series data of interest is the collection  $\{\mathbf{y}_t\}_{t=1}^T$ , where  $\mathbf{y}_t \in \mathbb{R}^K$ . In general, such time series can be conceptualized as a realization of the following process:

$$\mathbf{y}_t = f_t\left(\{\mathbf{y}_k\}_{k < t}, \quad \boldsymbol{\zeta}_t\right) \quad (1)$$

where the observation at time  $t$  depends on the process at pervious timepoints and some random innovation  $\boldsymbol{\zeta}_t$ . In order to keep our discussion relatively contained, we will assume that the time-series data is measured at approximately equal intervals.

In practice, this general process is often approximated by a Vector Autoregressive Model of order  $p$  (i.e. VAR(p)):

$$\mathbf{y}_t = \mathbf{v} + \boldsymbol{\Phi}_1 \mathbf{y}_{t-1} + \dots + \boldsymbol{\Phi}_p \mathbf{y}_{t-p} + \boldsymbol{\zeta}_t \quad (2)$$

In the above,  $\mathbf{v} = (\mathbf{I} - \boldsymbol{\Phi}_1 - \dots - \boldsymbol{\Phi}_p)\boldsymbol{\mu}$  where  $\boldsymbol{\mu}$  is the process mean. Without loss of generality, let us assume that  $\boldsymbol{\mu} = \mathbf{0}$  so that the intercept term  $\mathbf{v}$  can be omitted from (2). The zero-mean innovation process  $\boldsymbol{\zeta}_t$  is assumed to be uncorrelated across time, but is allowed to exhibit contemporaneous correlations. That is,  $\mathbb{E}[\boldsymbol{\zeta}_t] = \mathbf{0}$ ,  $\mathbb{E}[\boldsymbol{\zeta}_t \boldsymbol{\zeta}_t^\top] = \boldsymbol{\Sigma}_\zeta$  and  $\mathbb{E}[\boldsymbol{\zeta}_t \boldsymbol{\zeta}_j^\top] = \mathbf{0}_{K \times K}$  if  $t \neq j$ . Furthermore, we will make the standard assumption that the process of interest is weakly stationary (Hamilton, 1994). This is satisfied when the determinant of the lag-polynomial has no roots inside the unit circle:

$$\det\left(\mathbf{I} - \boldsymbol{\Phi}_1 z - \boldsymbol{\Phi}_2 z^2 - \dots - \boldsymbol{\Phi}_p z^p\right) \neq 0, \quad \forall |z| < 1, z \in \mathbb{C} \quad (3)$$

where  $\mathbb{C}$  is the complex plane (Kilian & Lütkepohl, 2017). If the raw data is not stationary (e.g.  $\mathbf{y}_t = \mathbf{d}_t + \tilde{\mathbf{y}}_t$  for some deterministic trend  $\mathbf{d}_t$ ), one can either use regression or

differencing to remove said trend and analyze either the residual or differenced time-series instead.

One key property of VAR models that is germane to our discussion is its Moving Average (MA) representation. Consider a VAR(1) model. By iterative substitution, we can write:

$$\mathbf{y}_t = \Phi_1 \mathbf{y}_{t-1} + \zeta_t \quad (4)$$

$$= \Phi_1 \left[ \Phi_1 \mathbf{y}_{t-2} + \zeta_{t-1} \right] + \zeta_t \quad (5)$$

$$= \Phi_1^2 \mathbf{y}_{t-2} + \Phi_1 \zeta_{t-1} + \zeta_t \quad (6)$$

$$= \dots \quad (7)$$

$$= \Phi_1^{h+1} \mathbf{y}_{t-h-1} + \sum_{i=0}^h \Phi_1^i \zeta_{t-i} \quad (8)$$

In the limit where  $h \rightarrow \infty$ , we have by the condition in (3) that  $\Phi_1^{h+1} \rightarrow \mathbf{0}$ . Hence, the stationary time-series  $\mathbf{y}_t$  has the following MA representation:

$$\mathbf{y}_t = \sum_{h=0}^{\infty} \Phi_1^h \zeta_{t-h} \quad (9)$$

where  $\Phi_1^h$  is the  $h^{th}$  MA coefficient matrix. A general VAR(p) model also has a MA representation. This can be seen by rewriting the VAR(p) model as a VAR(1) as follow:

$$\underbrace{\begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix}}_{=: Y_t} = \underbrace{\begin{bmatrix} \Phi_1 & \Phi_2 & \Phi_3 & \cdots & \Phi_{p-1} & \Phi_p \\ \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I} & \mathbf{0} \end{bmatrix}}_{=: \mathbf{C}} \underbrace{\begin{bmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \mathbf{y}_{t-3} \\ \vdots \\ \mathbf{y}_{t-p} \end{bmatrix}}_{=: Y_{t-1}} + \underbrace{\begin{bmatrix} \zeta_t \\ \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}}_{=: U_t} \quad (10)$$

where  $\mathbf{C}$  is sometimes called the ‘companion’ matrix (Kilian & Lütkepohl, 2017). The causality condition in (3) guarantees that  $\mathbf{C}^h$  converges to zero as  $h$  becomes large.

Accordingly, we have:

$$Y_t = \sum_{h=0}^{\infty} \mathbf{C}^h U_{t-h} \quad (11)$$

This MA representation is in terms of the augmented vector  $Y_t$ . To get the MA representation for the original data vector  $\mathbf{y}_t$ , let us define the following matrix:

$$\mathbf{J} = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \end{bmatrix} \quad (12)$$

Note that the following relations hold:

$$\mathbf{y}_t = \mathbf{J}Y_t, \quad U_t = \mathbf{J}^\top \mathbf{J}U_t, \quad \mathbf{J}U_t = \boldsymbol{\zeta}_t \quad (13)$$

Therefore,

$$\mathbf{y}_t = \mathbf{J}Y_t = \sum_{h=0}^{\infty} \mathbf{J}\mathbf{C}^h U_{t-h} = \sum_{h=0}^{\infty} \underbrace{\mathbf{J}\mathbf{C}^h \mathbf{J}^\top}_{=: \boldsymbol{\Psi}_h} \boldsymbol{\zeta}_{t-h} \quad (14)$$

and the corresponding MA coefficients for the VAR(p) process are denoted by  $\boldsymbol{\Psi}_h$ . Note that these MA coefficients are functions of the VAR parameters.

### Impulse Response Function

These MA coefficients are essential to the concept of impulse responses. In generic terms, impulse responses describe how the system under study responds to ‘shocks’ to the system. To be specific, let us consider rewriting the time index in (14). This yields:

$$\mathbf{y}_{t+h} = \sum_{k=0}^{\infty} \boldsymbol{\Psi}_k \boldsymbol{\zeta}_{t+h-k} \quad (15)$$

$$= \boldsymbol{\Psi}_0 \boldsymbol{\zeta}_{t+h} + \boldsymbol{\Psi}_1 \boldsymbol{\zeta}_{t+h-1} + \dots + \boldsymbol{\Psi}_{h-1} \boldsymbol{\zeta}_{t+1} + \boldsymbol{\Psi}_h \boldsymbol{\zeta}_t + \underbrace{\sum_{k=h+1}^{\infty} \boldsymbol{\Psi}_k \boldsymbol{\zeta}_{t+h-k}}_{g(\boldsymbol{\zeta}_{t-1}, \boldsymbol{\zeta}_{t-2}, \dots)} \quad (16)$$

where the last term is a function of the innovations before time  $t$ . Let

$\boldsymbol{\omega}_{t-1} = (\boldsymbol{\zeta}_{t-1}, \boldsymbol{\zeta}_{t-2}, \dots)$  denote the history of innovations before time  $t$ . Under this setup, we are interested in what occurs in the system in response to a one-time shock at time  $t$ . Specifically, the quantity of interest is the expected behavior of the variables at horizon  $h$ , conditional on the history  $\boldsymbol{\omega}_{t-1}$  prior to time  $t$ , in response to a shock at time  $t$  only and with all other shocks after time  $t$  set to zero (Koop et al., 1996). Formally, consider a shock  $\boldsymbol{\xi}^\top = (\xi^{(1)}, \xi^{(2)}, \dots, \xi^{(K)})$  where each element is some value in  $\mathbb{R}$ . The Impulse Response

Function (IRF) evaluated at horizon  $h$  corresponds to a contrast:

$$\text{IRF}(h, \boldsymbol{\xi}) \tag{17}$$

$$= \mathbb{E}[\mathbf{y}_{t+h} \mid \boldsymbol{\zeta}_{t+h} = \boldsymbol{\zeta}_{t+h-1} = \dots = \boldsymbol{\zeta}_{t+1} = \mathbf{0}, \boldsymbol{\zeta}_t = \boldsymbol{\xi}, \boldsymbol{\omega}_{t-1}] \tag{18}$$

$$- \mathbb{E}[\mathbf{y}_{t+h} \mid \boldsymbol{\zeta}_{t+h} = \boldsymbol{\zeta}_{t+h-1} = \dots = \boldsymbol{\zeta}_{t+1} = \mathbf{0}, \boldsymbol{\zeta}_t = \mathbf{0}, \boldsymbol{\omega}_{t-1}] \tag{19}$$

$$= \boldsymbol{\Psi}_h \boldsymbol{\xi} \tag{20}$$

The above simplified expression can be obtained because of the linearity in (16) and the zeros imposed on the shocks at time not prior to  $t$ . The term that involves the history of innovations before time  $t$  cancel out. Thus, we see that the IRF is encoded by the MA coefficients of a VAR(p) model (Kilian & Lütkepohl, 2017; Pesaran & Shin, 1998). Tracing this behavior across the horizons  $h = 0, 1, 2, \dots$  provides us with the IRF as a function of the horizon  $h$ . In substantive terms, the IRF describes how a one-time perturbation to the system affects the trajectory of our variables in the future.

Before we continue with the technical discussion, it is important to highlight that this abstraction has substantive applications within psychology. For one, consider the simple AR(1) model that is widely used in psychology (Jongerling et al., 2015; Liu, 2017). Specifically, the AR(1) model with white-noise innovations can be stated as:

$$y_t = \phi y_{t-1} + \zeta_t, \quad \zeta_t \sim \text{WN}(0, \sigma^2) \tag{21}$$

where  $|\phi| < 1$ . When the autoregressive coefficient  $\phi$  is positive, it is common to interpret it as a measure of ‘inertia’ (Jongerling et al., 2015). For values of  $\phi$  close to 0, the resulting time-series appears like random noise and fluctuates rapidly. In contrast, as  $\phi \rightarrow 1$ , the time-series becomes increasingly ‘clumpy’ as values change slowly due to the strong autoregressive effects. Thus, the parameter  $\phi$  captures the persistence of the system. However, we can also interpret such persistence in terms of the IRF. In this simple case, its IRF is simply equal to

$$\text{IRF}(h) = \phi^h \tag{22}$$

If  $\phi > 0$ , then the half-life of the IRF (i.e. the horizon  $h$  at which the the value of the IRF decays to half its original value) is the solution to

$$\phi^h = \frac{1}{2}\phi^0 \quad (23)$$

$$(\Rightarrow) \quad h \log \phi = -\log(2) \quad (24)$$

$$(\Rightarrow) \quad h = \frac{-\log(2)}{\log \phi} \quad (25)$$

which is an increasing function of  $\phi$  in the interval  $(0, 1)$ . As  $\phi \rightarrow 0$ , the half-life approaches zero. Conversely, as  $\phi \rightarrow 1$ , the half-life approaches infinity. Thus, inertia can also be interpreted in terms of the number of time steps needed for the IRF to decay to half its value over time. The main advantage of viewing inertia through the lens of IRFs is that it is not model dependent. The half-life of the IRF can be computed for general ARIMA models (Koop et al., 1996), whereas the interpretation of  $\phi$  as inertia only holds for AR(1) models with positive autoregressive coefficients. Such an approach is routinely used within econometrics (Seong et al., 2006), but is rarely seen within psychology.

There is also substantial interest in impulse responses in multivariate systems. For instance, researchers have used impulse response analysis to probe how emotional states can impact immune system activity (Seizer et al., 2023), assessed if physical activity elicits downstream changes in affect (F. Bos et al., 2018) and have also investigated the long-run equilibrium of an impulse response in emotional dynamics (Yang et al., 2018). Moreover, studies have also built upon the idea of quantifying inertia in a multivariate system using impulse responses. This has been done by approximating the number of timesteps needed for the impulse response to return arbitrarily close to baseline (Lydon-Staley, Leventhal, et al., 2021; Yang et al., 2018). Throughout these applications, the main idea is that the analysis of impulse responses can quantify the reverberating effects of feedback loops in a multivariate system (Lydon-Staley, Cornblath, et al., 2021), whereas a direct inspection of the lagged relations do not immediately yield such insight. The IRFs also retain the same interpretation regardless of the order  $p$  in a VAR( $p$ ) model. This is unlike the direct

interpretation of the  $p$  lagged matrices, which becomes increasingly difficult when  $p$  is not small. Moreover, scalar quantities obtained from IRFs can also be represented as networks (F. Bos et al., 2018), from which network measures can be extracted for further analysis.

These varied applications suggest that impulse response analysis should reveal important insights into the behavior of multivariate systems. However, the study of impulse responses in a VAR come with certain complications. The most immediate problem lies in the choice of the shock vector  $\boldsymbol{\xi}$  in (20). Most oftenly, we want to apply a shock to a single variable in the system in isolation (F. Bos et al., 2018; Lydon-Staley, Leventhal, et al., 2021; Seizer et al., 2023; Yang et al., 2018, 2019). That is, we want to consider  $\boldsymbol{\xi}_j = (0, \dots, 0, \sigma_j, 0, \dots, 0)$  where the shock is applied only to the  $j^{th}$  variable and  $\sigma_j$  is the standard deviation of the  $j^{th}$  innovation. Shocks of this form are sometimes called ‘structural’ or ‘orthogonal’ shocks (Kilian & Lütkepohl, 2017; Sims, 1980). These shocks are often of most substantive interest as they can naturally inform real-world intervention analyses — all else being equal, we want to understand the changes caused to a system by an isolated shock to a given variable. This natural choice of a shock is, however, usually not empirically meaningful. The main issue is that the innovations are typically correlated with covariance  $\boldsymbol{\Sigma}_\zeta$  and orthogonal shocks supplied by setting  $\boldsymbol{\zeta}_t = \boldsymbol{\xi}_j$  do not conform with the data generating process (Koop et al., 1996). As an extreme example, consider  $\boldsymbol{\xi} = (\sigma_1, 0)$  but with the correlation between the bivariate shocks set to 0.99. Such a shock will almost never be observed based on the data-generating process; the application of such shocks is akin to extrapolating beyond the range of the data in regression. The IRFs computed using such unwarranted orthogonal shocks are misleading at best. In general, we need to take the observed covariance of innovations  $\boldsymbol{\Sigma}_\zeta$  into account when analyzing the effect of shocks.

### Generalized Impulse Response Function

There are two ways in which we can resolve this conundrum. The first is to completely sidestep the study of structural shocks and restrict ourselves to shocks that are compatible with  $\boldsymbol{\Sigma}_\zeta$ . The idea is to abandon the definition in (20) and instead consider a



‘generalized’ Impulse Response Function (gIRF) (Koop et al., 1996; Pesaran & Shin, 1998). This is somewhat of a misnomer since the structural IRF is not subsumed under the gIRF. The gIRF for a shock to the  $j^{th}$  variable is given by:

$$\text{gIRF}(h)^{(j)} = \mathbb{E}[\mathbf{y}_{t+h} \mid \boldsymbol{\zeta}_t^{(j)} = v, \boldsymbol{\omega}_{t-1}] - \mathbb{E}[\mathbf{y}_{t+h} \mid \boldsymbol{\omega}_{t-1}] \quad (26)$$

where  $\boldsymbol{\zeta}_t^{(j)}$  is the  $j^{th}$  element of the innovation vector at time  $t$ . We will further assume that the innovations are not only uncorrelated across time, but are also independent:

$\boldsymbol{\zeta}_t \perp \boldsymbol{\zeta}_j, t \neq j$ . Using the MA representation in (16) and the fact that the innovations have mean zero, we have

$$\text{gIRF}(h)^{(j)} = \boldsymbol{\Psi}_h \mathbb{E}[\boldsymbol{\zeta}_t \mid \boldsymbol{\zeta}_t^{(j)} = v] + \left( \sum_{k=h+1}^{\infty} \boldsymbol{\Psi}_k \boldsymbol{\zeta}_{t+h-k} \right) \quad (27)$$

$$- \left( \sum_{k=h+1}^{\infty} \boldsymbol{\Psi}_k \boldsymbol{\zeta}_{t+h-k} \right) \quad (28)$$

$$= \boldsymbol{\Psi}_h \mathbb{E}[\boldsymbol{\zeta}_t \mid \boldsymbol{\zeta}_t^{(j)} = v] \quad (29)$$

When the innovations are assumed to be multivariate gaussian, this conditional expectation has the explicit form (Koop et al., 1996):

$$\mathbb{E}[\boldsymbol{\zeta}_t \mid \boldsymbol{\zeta}_t^{(j)} = v] = \frac{1}{\sigma_j^2} \boldsymbol{\Sigma}_{\boldsymbol{\zeta}} \mathbf{e}_j v \quad (30)$$

where  $\mathbf{e}_j$  is equal to unity in the  $j^{th}$  coordinate and zero elsewhere. Hence, a shock to the  $j^{th}$  variable with magnitude equal to one standard deviation of its corresponding innovation variance is given by:

$$\text{gIRF}(h)^{(j)} = \frac{\boldsymbol{\Psi}_h \boldsymbol{\Sigma}_{\boldsymbol{\zeta}} \mathbf{e}_j}{\sigma_j} \quad (31)$$

In other words, we circumvent the issue of non-compatible orthogonal shocks by considering innovations that would have been generated by the data-generating process. This same framework applies even when the innovations are not multivariate gaussian, but the resulting expression would be more complicated (Koop et al., 1996).

There are several important distinctions to be made regarding the gIRF and the structural IRF. For one, the definition in (26) considers what happens to the variables at

horizon  $h$  upon observing an innovation for variable  $j$ . It does not force the innovations of the other variables to be zero (or any other value at all); it simply integrates over them. Unlike the structural IRF, it does not place constraints on the shocks from time  $t + 1$  till  $t + h$ . It thus does not capture a ‘pure’ response to a shock to variable  $j$  at time  $t$ . One can think of the gIRF and the IRF as capturing ‘observation’ and ‘intervention’ respectively (Pearl, 2009). In the former, we merely consider what the expected response would be if we were to observe a shock at time  $t$ . In contrast, the latter is concerned with the expected response if we were to surgically intervene in the system and induce a shock only at time  $t$ , while somehow removing the influence of all external sources from  $t + 1$  to  $t + h$ . Despite its name, the former clearly provides a much weaker form of inference than the latter.

As a side note, the gIRF can be computed as a function of estimated VAR parameters. Specifically, the lagged VAR parameters can be consistently estimated via Ordinary Least Squares:

$$\arg \min_{\Phi_1, \dots, \Phi_p} \sum_{t=p+1}^T \|\mathbf{y}_t - \Phi_1 \mathbf{y}_{t-1} - \dots - \Phi_p \mathbf{y}_{t-p}\|_2^2 \quad (32)$$

where  $T$  is the length of the time-series. These lagged parameters can be used to form the relevant moving average matrices. Similarly, a consistent estimator of the innovation covariance  $\hat{\Sigma}_\zeta$  can be obtained from the least square residuals (Kilian & Lütkepohl, 2017). Other estimation approaches are also possible (Gates & Molenaar, 2012), but OLS is the most straightforward.

### Structural Vector Autoregressive Models

Given the weak interpretation afforded by the gIRF, one might wonder if there is a way to estimate an IRF with a structural interpretation. Such a structural interpretation is often desired in psychological practice. For example, given resting-state fMRI data from one person, a researcher might be interested in inferring how stimulation to parts of the cortex can alter neural activity in subcortical regions without actually having to conduct said stimulation. If feasible, this would be very useful for personalized medicine since we

can make a reasonable guess at the effects of intervention even before applying the intervention itself. Similarly, a clinician with access to a person's daily diary data might be interested in the effects of physical activity on depressive symptoms for that person. However, this interest usually does not stop at merely observing that physical activity has some relation to depression. Rather, the primary interest here is in the causal impact of increased physical activity (perhaps induced by providing gym vouchers) on the subsequent evolution of depressive symptoms over time. Again, cost savings would be substantial if we could somehow infer the effect of said intervention purely from the observational time series and only apply such interventions to individuals for whom the intervention is inferred to be somewhat effective. It turns out that this is possible under certain (stringent) assumptions which we will explicate below.

The main framework under which such structural impulses can be obtained is through the Structural Vector Autoregressive model (SVAR). There are several equivalent ways of specifying an SVAR. One common representation used in psychology (Gates et al., 2010; Molenaar, 2017) is the following:

$$(\mathbf{I} - \mathbf{A}_0)\mathbf{y}_t = \mathbf{A}_1\mathbf{y}_{t-1} + \dots + \mathbf{A}_p\mathbf{y}_{t-p} + \boldsymbol{\delta}_t \quad (33)$$

$$\mathbb{E}[\boldsymbol{\delta}_t\boldsymbol{\delta}_t^\top] = \text{diag}(\sigma_{\delta_1}^2, \sigma_{\delta_2}^2, \dots, \sigma_{\delta_K}^2) \quad (34)$$

with the structural innovations  $\boldsymbol{\delta}_t$  having mean zero and are also uncorrelated across time. The key difference here is that the covariance of  $\boldsymbol{\delta}_t$  is diagonal. We will assume that  $\mathbf{A}_0$  has zero on its diagonals and that  $(\mathbf{I} - \mathbf{A}_0)^{-1}$  exists. The matrices  $(\mathbf{A}_0, \mathbf{A}_1, \dots, \mathbf{A}_p)$  and the variances  $\sigma_{\delta_j}^2$  are the structural VAR parameters. One way to interpret this representation is that the contemporaneous relations among the innovations are now absorbed into the directed contemporaneous matrix  $\mathbf{A}_0$ . A variable at time  $t$  is regressed not only on lagged versions of the variables in the system, but also on the other variables measured at the same timepoint  $t$ . Another representation that is more common in econometrics is the

following (Kilian & Lütkepohl, 2017; Lanne et al., 2010):

$$\mathbf{B}_0 \mathbf{y}_t = \mathbf{B}_1 \mathbf{y}_{t-1} + \dots + \mathbf{B}_p \mathbf{y}_{t-p} + \tilde{\boldsymbol{\delta}}_t \quad (35)$$

$$\mathbb{E}[\tilde{\boldsymbol{\delta}}_t \tilde{\boldsymbol{\delta}}_t^\top] = \mathbf{I} \quad (36)$$

where the matrix  $\mathbf{B}_0$  is unconstrained. These two representations are interchangeable. To go from the former to the latter, define  $\boldsymbol{\Lambda}_\delta = \text{diag}(\sigma_{\delta_1}, \sigma_{\delta_2}, \dots, \sigma_{\delta_K})$ ,  $\boldsymbol{\delta}_t := \boldsymbol{\Lambda}_\delta \tilde{\boldsymbol{\delta}}_t$  and set  $\mathbf{B}_0 := \boldsymbol{\Lambda}_\delta^{-1}(\mathbf{I} - \mathbf{A}_0)$  with  $\mathbf{B}_l = \boldsymbol{\Lambda}_\delta^{-1} \mathbf{A}_l$  for  $l \in \{1, \dots, p\}$ . Conversely, let  $\boldsymbol{\Lambda}_B := \text{diag}(\mathbf{B}_0)$ . Define  $\mathbf{Q} := \boldsymbol{\Lambda}_B^{-1} \mathbf{B}_0$ , and set  $\mathbf{A}_l = \boldsymbol{\Lambda}_B^{-1} \mathbf{B}_l$ . Moreover, let  $\boldsymbol{\delta}_t := \boldsymbol{\Lambda}_B^{-1} \tilde{\boldsymbol{\delta}}_t$  such that its covariance is diagonal. By construction,  $\mathbf{Q}$  has a unit diagonal and can thus be written in the form of  $(\mathbf{I} - \mathbf{A}_0)$ . In the following sections, we will use one of these two representations depending on the use case.

More importantly, the parameters in the structural representation in (33) to (36) are different from those in the VAR stated in (2). The matrices specified in (2) and the corresponding innovation covariance matrix  $\boldsymbol{\Sigma}_\zeta$  are termed the reduced-form VAR parameters (Kilian & Lütkepohl, 2017; Rubio-Ramírez et al., 2010). In contrast, the SVAR parameters are typically called the ‘structural’ parameters. The two sets of parameters are related — we can always transform an SVAR into a reduced-form VAR (Gates et al., 2010; Molenaar, 2017; Rubio-Ramírez et al., 2010). Using (33), we can multiply both sides by  $(\mathbf{I} - \mathbf{A}_0)^{-1}$  to obtain:

$$\mathbf{y}_t = (\mathbf{I} - \mathbf{A}_0)^{-1} \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + (\mathbf{I} - \mathbf{A}_0)^{-1} \mathbf{A}_p \mathbf{y}_{t-p} + \underbrace{(\mathbf{I} - \mathbf{A}_0)^{-1} \boldsymbol{\delta}_t}_{=:\boldsymbol{\zeta}_t} \quad (37)$$

$$\mathbb{E}[\boldsymbol{\zeta}_t \boldsymbol{\zeta}_t^\top] = \left[ (\mathbf{I} - \mathbf{A}_0)^{-1} \right] \boldsymbol{\Lambda}_\delta^2 \left[ (\mathbf{I} - \mathbf{A}_0)^{-1} \right]^\top =: \boldsymbol{\Sigma}_\zeta \quad (38)$$

such that  $\boldsymbol{\Phi}_l = (\mathbf{I} - \mathbf{A}_0)^{-1} \mathbf{A}_l$  for  $l \in \{1, \dots, p\}$ . From (37), we can see that an SVAR amounts to a linear decomposition of the reduced-form innovation terms  $\boldsymbol{\zeta}_t$  into a vector of uncorrelated structural innovations  $\boldsymbol{\delta}_t$ .

The most important aspect of an SVAR is that each element in the structural innovation vector  $\boldsymbol{\delta}_t$  is uncorrelated with each other. This implies that (1) the shocks can

be specified in isolation of one another and (2) any shock vector  $\boldsymbol{\xi} \in \mathbb{R}^K$  could (up to scaling) be specified without contradicting the data-generating process. Accordingly, this suggests that we specify our IRF in terms of the structural, rather than the reduced-form, innovations:

$$\text{IRF}(h, \boldsymbol{\xi}) \tag{39}$$

$$= \mathbb{E}[\mathbf{y}_{t+h} \mid \boldsymbol{\delta}_{t+h} = \boldsymbol{\delta}_{t+h-1} = \dots = \boldsymbol{\delta}_{t+1} = \mathbf{0}, \boldsymbol{\delta}_t = \boldsymbol{\xi}, \boldsymbol{\omega}_{t-1}] \tag{40}$$

$$- \mathbb{E}[\mathbf{y}_{t+h} \mid \boldsymbol{\delta}_{t+h} = \boldsymbol{\delta}_{t+h-1} = \dots = \boldsymbol{\delta}_{t+1} = \mathbf{0}, \boldsymbol{\delta}_t = \mathbf{0}, \boldsymbol{\omega}_{t-1}] \tag{41}$$

$$= \boldsymbol{\Psi}_h(\mathbf{I} - \mathbf{A}_0)^{-1} \boldsymbol{\xi} \tag{42}$$

where  $\boldsymbol{\omega}_t$  is interpreted as the history of structural innovations up to time  $t$ . Note that the moving average matrices here are expressed in terms of  $\boldsymbol{\Phi}_l$  as in (14).

Clearly, computation of the structural IRF is a trivial exercise if we knew the values in  $\mathbf{A}_0$ : we could simply estimate the reduced-form lagged matrices using OLS, form the moving average matrices and multiply the corresponding moving average matrix with  $(\mathbf{I} - \mathbf{A}_0)^{-1} \boldsymbol{\xi}$ . This highlights that the critical issue in SVAR modeling lies in the estimation of  $\mathbf{A}_0$  (or  $\mathbf{B}_0$ ). However, it turns out that this is a highly non-trivial problem. The main obstacle is that an infinite number of SVAR models can yield the same reduced-form model due to rotational freedom (Kilian & Lütkepohl, 2017; Molenaar, 2017). To demonstrate this, it is easier if we work with the rescaled representation in (35) and (36). Let  $\mathbf{P}$  be an orthogonal rotation matrix such that  $\mathbf{P}^\top \mathbf{P} = \mathbf{P} \mathbf{P}^\top = \mathbf{I}$ . By pre-multiplying (35) with  $\mathbf{P}$ , we obtain another SVAR:

$$\underbrace{\mathbf{P} \mathbf{B}_0}_{=: \mathbf{B}_0^*} \mathbf{y}_t = \underbrace{\mathbf{P} \mathbf{B}_1}_{=: \mathbf{B}_1^*} \mathbf{y}_{t-1} + \dots + \underbrace{\mathbf{P} \mathbf{B}_p}_{=: \mathbf{B}_p^*} \mathbf{y}_{t-p} + \underbrace{\mathbf{P} \tilde{\boldsymbol{\delta}}_t}_{=: \tilde{\boldsymbol{\delta}}_t^*} \tag{43}$$

$$\mathbb{E}[\tilde{\boldsymbol{\delta}}_t^* (\tilde{\boldsymbol{\delta}}_t^*)^\top] = \mathbf{P} \mathbf{P}^\top = \mathbf{I} \tag{44}$$

with new structural parameters  $\mathbf{B}_0^*, \mathbf{B}_1^*, \dots, \mathbf{B}_p^*$ . However, we observe that the corresponding reduced-form VAR has lagged parameters

$\boldsymbol{\Phi}_l^* = (\mathbf{B}_0^*)^{-1} \mathbf{B}_l^* = \mathbf{B}_0^{-1} \mathbf{P}^\top \mathbf{P} \mathbf{B}_l = \mathbf{B}_0^{-1} \mathbf{B}_l = \boldsymbol{\Phi}_l$  and the reduced-form innovation

covariance similarly has the form  $\Sigma_{\zeta}^* = (\mathbf{B}_0^*)^{-1}[(\mathbf{B}_0^*)^{-1}]^\top = \mathbf{B}_0^{-1}[(\mathbf{B}_0)^{-1}]^\top = \Sigma_{\zeta}$ . In other words, we have two SVARs that produce the same reduced-form VAR. Since  $\mathbf{P}$  is an arbitrary orthogonal matrix, there is an infinite set of SVARs that correspond to a given reduced-form VAR. We thus have entire sets of SVARs related via rotation that are observationally equivalent (Rubio-Ramírez et al., 2010). Although we can consistently estimate the reduced-form VAR using OLS, it is not necessarily clear how one should choose from one of the many equivalent SVAR representations. At this point, it should be emphasized that the main point of an SVAR analysis is interpretational in nature — we want to understand the actual impact of orthogonal shocks to the system. The SVAR itself does not confer any advantages in terms of forecasting as compared to the vanilla VAR (Kilian & Lütkepohl, 2017). This should be obvious since an SVAR can always be written as a reduced-form VAR, so a VAR is all that is needed for prediction purposes (at least when nuances regarding estimation are not considered).

In the following sections, we will review existing strategies that have been used to disambiguate between the rotationally equivalent SVARs. The strategies we review can be categorized into three types: zero restrictions, data-driven approaches and sign restrictions. For each strategy, we will consider its merits and shortcomings while also considering the validity of said strategy in the context of psychology. With some abuse of notation, we will refer to  $\mathbf{A}_0$  and  $\mathbf{B}_0$  somewhat interchangeably by appealing to their equivalence in representing the SVAR model as discussed earlier.

### ***Identification via Zero Restrictions***

One way to identify the SVAR model parameters is to impose structural zeros in either the  $\mathbf{B}_0^{-1}$  or  $\mathbf{B}_0$  matrices. Since these two matrices describe the contemporaneous relations among the shocks and variables, restrictions on these matrices are also termed ‘short-run’ restrictions (Kilian & Lütkepohl, 2017). These zero restrictions are most often imposed through theoretical considerations. For example, the matrix  $\mathbf{B}_0^{-1}$  can be interpreted as a impact multiplier matrix that maps the structural innovations to the

variables in the system. One could then impose the restriction that a certain shock has no contemporaneous on some variables. As a concrete example, we might believe that a shock applied to the visual cortex has no contemporaneous effect on neural activity in deep subcortical structures.

Traditionally, one approach to imposing zero restrictions is to assume that the variables in the VAR satisfy a recursive ordering such that the  $\mathbf{B}_0^{-1}$  matrix is lower triangular (Sims, 1980). When such a lower triangular structure is known, we can perform a cholesky decomposition on the estimated reduced-form innovation covariance matrix to obtain  $\hat{\Sigma}_\zeta = \mathbf{L}\mathbf{L}^\top$ . Then we can simply set  $\mathbf{B}_0^{-1} = \mathbf{L}$  and invert it accordingly. This decomposition is unique if the diagonals of  $\mathbf{L}$  are constrained to be positive (i.e. each shock has non-zero contemporaneous impact on its associated variable). While straightforward, this recursive approach has one major failing: in practice, we rarely have enough information to specify a recursive ordering, if it even exists in the first place. Despite this, this approach has been used in applied work within psychology where the researchers estimated the IRFs by trying different recursive orderings of the variables (Seizer et al., 2023). This is, however, not credible since we achieve identification by claiming knowledge of the ordering of variables, but at the same time claim to be unsure as to what the actual ordering is (Kilian & Lütkepohl, 2017).

A more generic approach to identifying the SVAR is to examine the covariance relation in (38). This relation essentially specifies an SEM model for the covariance of the reduced-form innovations (Gates et al., 2010). Accordingly, the SVAR model is also sometimes termed as a ‘structural causal model for the error’ terms (Swanson & Granger, 1997). If a sufficient number of structural zeros are available in  $\mathbf{A}_0$  to render the corresponding SEM identifiable, then estimation can be done by finding parameters in  $\mathbf{A}_0$  that minimizes some discrepancy function between the model-implied covariance matrix and the estimated reduced-form innovation covariance matrix  $\hat{\Sigma}_\zeta$ . In general, a necessary order condition is that we need at least  $K(K - 1)/2$  restrictions on  $\mathbf{A}_0$  (Kilian & Lütkepohl,

2017; Rubio-Ramírez et al., 2010). Again, this might be unrealistic in practice, especially if the number of variables  $K$  is not small. It should be noted that zero restrictions can also be imposed on the long-run expected response of a shock (i.e.  $\text{IRF}^{(j \rightarrow m)}(\infty) = 0$  for some shock  $j$  and variable  $m$ ) to help achieve identification. Sufficient algebraic conditions are also available to determine if some combination of short-run and long-run zero restrictions can identify a model (Rubio-Ramírez et al., 2010). Nevertheless, psychological theories are rarely sufficiently mature for unambiguous zero restrictions to be imposed. This limits the practical use of exact zero restrictions in the identification of SVARs in psychology.

### ***Data-Driven Approaches***

**Automated Model Search.** Instead of pursuing exact identification, a large body of work in the SVAR literature has instead focused on automated procedures that attempt to recover the true model when the true SVAR model is sparse (Beltz & Molenaar, 2016; Demiralp et al., 2008; Duffy et al., 2021; Gates & Molenaar, 2012; Gates et al., 2010; Swanson & Granger, 1997; Weigard et al., 2023). That is, the true contemporaneous matrix  $\mathbf{A}_0$  is assumed to contain many exact zeros, but the location of these zeros are unknown. Usually, the pattern of zeros in  $\mathbf{A}_0$  is further assumed to correspond to an acyclic graph when the matrix is viewed as an adjacency matrix among the variables (Beltz & Molenaar, 2016; Demiralp et al., 2008; Gates et al., 2010). In other words, we have a recursive model, but the recursive ordering is unknown. The key idea here is that the excess zero restrictions in  $\mathbf{A}_0$  often have testable implications, and hence the data is informative about the structure of the model (Demiralp et al., 2008; Swanson & Granger, 1997). That is, we assume that the true data generating model is overidentified. For a given candidate  $\mathbf{A}_0$ , these implications often take the form of partial correlation constraints or tetrad difference constraints (Swanson & Granger, 1997). If the observed correlations / tetrads violate these constraints, the candidate  $\mathbf{A}_0$  is rejected. Automated model search procedures take this process one step further by iteratively searching for models that exhibit minimal violations of these constraints.



There are a variety of such search algorithms. Within the econometric literature, causal search methods based on Directed Acyclic Graphs have been applied to the reduced-form innovations to find a permissible  $\mathbf{A}_0$ . Simulation results have shown that recovery of the true SVAR model can be fairly robust when the signal-to-noise ratio is high (Demiralp et al., 2008). A similar line of research has emerged within psychology under the term ‘Unified Structural Equation Modeling’ (uSEM) (Gates et al., 2010; Kim et al., 2007). This approach is slightly different in that a search algorithm is applied not only to the reduced-form innovations, but is instead applied to the actual time-series itself. The central idea is that one can rewrite the time-series data in block-toeplitz form

$$\underbrace{\begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \end{bmatrix}}_{\mathbf{Z}} = \begin{bmatrix} \mathbf{A}_0 & \mathbf{\Phi}_1 \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \underbrace{\begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \end{bmatrix}}_{\mathbf{Z}} + \underbrace{\begin{bmatrix} \boldsymbol{\delta}_t \\ \mathbf{y}_{t-1} \end{bmatrix}}_{\boldsymbol{\epsilon}} \quad (45)$$

for a VAR(1) model. The covariance of  $\boldsymbol{\epsilon}$  is often restricted such that

$$\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}} = \begin{bmatrix} \boldsymbol{\Lambda}_{\boldsymbol{\delta}}^2 & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_y(0) \end{bmatrix} \quad (46)$$

where  $\boldsymbol{\Sigma}_y(0)$  is the lag-0 autocovariance of the data  $\mathbf{y}_t$ . A similar expression is applicable to VAR(p) models. Written in this way, the above expression is a structural equation model for  $\mathbf{Z}$ . For any candidate choice of model parameters  $(\mathbf{A}_0, \mathbf{\Phi}_1, \boldsymbol{\Lambda}_{\boldsymbol{\delta}}^2, \boldsymbol{\Sigma}_y(0))$ , goodness-of-fit indices can then be used to quantify the degree of discrepancy between the model-implied and empirical covariances of the stacked data  $\mathbf{Z}$  (Gates et al., 2010). Model search under this SEM framework proceeds through the use of modification indices. The procedure is first initialized by starting with some ‘null’ model. For example, we might set  $\mathbf{A}_0 = \mathbf{0}$  and set  $\mathbf{\Phi}_1$  to be diagonal (Weigard et al., 2023). Thereafter, the modification indices are computed and the parameters (i.e. paths) with the highest modification index are chosen to be freed up. The model is then re-estimated with the additional path and this procedure iterates until a pre-specified stopping rule is satisfied. This stopping rule is typically based on established cut-offs for select goodness-of-fit indices (Gates & Molenaar,

2012). In general, the resulting SVAR cannot be rotated without destroying the pattern of zeros found in  $\widehat{\mathbf{A}}_0$ , thereby resulting in a unique estimate of the SVAR (Molenaar, 2017). This preference for sparsity helps pin down the SVAR of interest among the infinite set of SVARs compatible with the data.

Variations of this SEM-based model search procedure have also been developed. In perhaps what is the most popular variant in psychology, the Group Iterative Multiple Model Estimation (GIMME) algorithm (Gates & Molenaar, 2012) uses not just one time-series from one participant, but instead leverages multiple time-series collected from multiple persons. The key insight here is that the signal in each individual time-series may be weak, but these signals exhibit a certain degree of commonality among the multiple individuals. By pooling information across multiple time-series, we can thus improve the signal-to-noise ratio and thus improve the performance of model search algorithms. Under GIMME, the rule for selecting paths to be freed after the computation of the modification indices is also modified slightly: a path is freed for all persons if its modification index is statistically significant for the largest number of people, where significance is fixed at the 0.01 level (Gates & Molenaar, 2012). There is usually also a requirement that the modification index must be significant for more than 75% of the people for a path to be freed, although the exact proportion can be set by the researcher. GIMME also conducts further refinements by conducting individual-level model search after the initial group-level search. Other extensions have been further developed for dealing with edge cases where ties among the candidate paths to be freed are encountered (Beltz & Molenaar, 2016). Results from large-scale simulation studies have shown that GIMME can reliably recover the underlying SVAR model (Duffy et al., 2021; Gates & Molenaar, 2012; Nestler & Humberg, 2021).

The primary benefit of such model search procedures is that precise prior knowledge of  $\mathbf{A}_0$  is not required. This is very appealing from an applied perspective. If we believe that our data at hand is generated by a process similar to what was investigated in the

existing large-scale simulation studies, then we can be reasonably confident that an appropriate SVAR model is found. IRFs can then be computed in the usual way. In fact, most empirical works in psychology employing IRF in their analyses utilize the model search approach to estimate SVARs (Gunther et al., 2024; Lydon-Staley, Leventhal, et al., 2021; Yang et al., 2018, 2019).

However, the current practice of using model search algorithms in psychology can be problematic for several reasons. Firstly, the vast majority of simulation studies were conducted with extremely sparse  $\mathbf{A}_0$  matrices (Beltz & Molenaar, 2016; Demiralp et al., 2008; Gates & Molenaar, 2012; Gates et al., 2010; Weigard et al., 2023). Yet, there is often little justification for why such a belief should hold for SVAR processes in the real world. In fact, recovery of directionality (i.e. the location of zeros) from the data itself is not possible when the number of zeros in  $\mathbf{A}_0$  makes  $\mathbf{A}_0$  just-identified (Demiralp et al., 2008). There is also substantial evidence that recovery of the true model depends critically on the strength of the SVAR parameters. Simulation results demonstrate that recovery of the true model can be very poor when SVAR parameters are of relatively small magnitude (Demiralp et al., 2008; Weigard et al., 2023), as might be expected when studying day-by-day fluctuations in emotional time-series (Weigard et al., 2023). Moreover, model recovery is far from perfect even when the parameter magnitudes are fairly high (Weigard et al., 2023). In general, model search procedures do not guarantee point identification of the SVAR parameters, although a unique estimate may be returned by the search algorithm due to idiosyncratic stopping rules. The existing simulations using GIMME have also almost always been conducted using a VAR(1) (Beltz & Molenaar, 2016; Gates & Molenaar, 2012; Weigard et al., 2023), even though empirical evidence suggests that lag orders greater than 1 might be present in psychological applications (Weigard et al., 2023). The implications of such imperfect recovery and misspecified lag order on the subsequent accuracy of the computed IRF is largely understudied.

***Other (Generic) Limitations.*** The limitations considered in this section are common to data-driven approaches, including the ones that we will discuss in later sections. These concerns are not unique to the model search paradigm. We state these limitations here for convenience and ease of reading. We will reference these limitations again when discussing other data-driven approaches.

One important generic concern is the assumption of acyclicity. This is often made in the aforementioned simulation studies. For a matrix  $\mathbf{A}_0$  satisfying acyclicity, we can always permute the variables such that the resulting matrix  $\tilde{\mathbf{A}}_0$  (and hence  $\tilde{\mathbf{B}}_0$ ) is lower-triangular. This implies that the impact multiplier matrix  $(\tilde{\mathbf{B}}_0)^{-1}$  is lower triangular as well, which necessarily means that certain shocks to the system have exactly zero contemporaneous effects on some of the variables. While this might be possible in a dynamic system governed by physical constraints such as in fMRI, it is not immediately obvious why this should hold when studying abstract concepts such as emotions. For example, in studying a bivariate system consisting of physical activity and positive affect, it is unclear why a shock to either variable should have zero effect on the other within the same time window.

Another criticism is that a loose interpretation of ‘shocks’ that correspond to an estimated  $\mathbf{A}_0$  matrix can be problematic for psychology. To see why this might be an issue, it is best to compare this with the prevailing practice in other fields. In the econometrics literature, it is often emphasized that a distinction must be made between the ‘statistical identification’ and the ‘economical identification’ of shocks (Kilian & Lütkepohl, 2017). That is, while the matrix  $\mathbf{A}_0$  may be identified through mathematical means, that does not automatically entail that the shocks are readily interpretable in substantive terms. A clear defensible rationale must be given to label a given shock. For example, consider the following canonical example in economics (Kilian & Lütkepohl, 2017):

$$\begin{bmatrix} (\text{Oil Price})_t \\ (\text{Oil Availability})_t \end{bmatrix} = \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} + \underbrace{\begin{bmatrix} + & - \\ + & + \end{bmatrix}}_{\mathbf{B}_0^{-1}} \begin{bmatrix} \Delta(\text{Oil Demand})_t \\ \Delta(\text{Oil Supply})_t \end{bmatrix} \quad (47)$$

where we have only specified the signs of the impact multiplier matrix. The two variables that make up our data are oil price and the availability of oil at time  $t$ . However, the corresponding shocks are not simply labelled as the ‘shock to oil price’ or as the ‘shock to oil availability’. Instead, an analyst specifies the nature of the shock based on external economic considerations (i.e. supply and demand), which in turn impose restrictions on the signs of the impact multiplier matrix. Specifically, a demand shock drives up the price but will also provide incentive for companies to increase production. Conversely, a positive supply shock reduces the price but increases the availability of oil. That is, the labelling of the shocks informs the matrix  $\mathbf{B}_0^{-1}$ . This is also the logic behind the imposition of zero restrictions on  $\mathbf{B}_0^{-1}$  discussed in the previous section. However, the use of model search procedures implies that we need to label the shocks post-hoc. This can lead to issues with interpretation when considering time-series from multiple participants. For example, consider a trivariate system with physical activity, negative affect and positive affect. Our model search might return:

$$\text{Person A: } \begin{bmatrix} (\text{Physical Activity})_t \\ (\text{Negative Affect})_t \\ (\text{Positive Affect})_t \end{bmatrix} = \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} + \begin{bmatrix} + & * & * \\ + & * & * \\ - & * & * \end{bmatrix} \begin{bmatrix} (\text{'Physical Activity Shock'})_t \\ * \\ * \end{bmatrix} \quad (48)$$

$$\text{Person B: } \begin{bmatrix} (\text{Physical Activity})_t \\ (\text{Negative Affect})_t \\ (\text{Positive Affect})_t \end{bmatrix} = \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} + \begin{bmatrix} + & * & * \\ - & * & * \\ + & * & * \end{bmatrix} \begin{bmatrix} (\text{'Physical Activity Shock'})_t \\ * \\ * \end{bmatrix} \quad (49)$$

In the above, we have denoted terms that are not of central interest with  $*$ . Although the first ‘shock’ for both persons are related to physical activity, they can have very different substantive meanings. For example, the shock for person A might correspond to increased exposure to fat shaming, whereas the shock for person B might be due to increased activity

from peer group effects. Although both increase physical activity, they have effects of opposite polarity for affect. The issue with current practice in psychology is that shocks with the same position in the VAR system are often treated as representing the same thing across people without regard to the actual signs of the impact multiplier matrix.

Downstream analyses then use some summary statistics of the resulting IRFs to correlate with person characteristics (Gunther et al., 2024; Lydon-Staley, Leventhal, et al., 2021; Yang et al., 2018). It is not clear how meaningful such downstream analyses are when the shocks might differ in their empirical content. This is akin to conducting individual-specific factor analysis on each person’s time series data (Hamaker et al., 2005) and then, without inspecting the factor loadings, labeling the first factor across people as the same latent variable. In general, there is no reason why these shocks resulting from model search should automatically have the same substantive meaning.

**Independent Component Analysis.** A different approach to SVAR analysis is through the lens of Independent Component Analysis (ICA) (Hyvarinen et al., 2010; Moneta et al., 2013). The ICA approach is similar in spirit to the model search paradigm — instead of imposing restrictions based on theory, we turn to properties of the data for additional insight. Where ICA differs is that it explicitly assumes that the structural innovations are non-gaussian and independent. When these assumptions are satisfied, the relation between the reduced-form and structural innovations is essentially an ICA model:

$$\zeta_t = (\mathbf{I} - \mathbf{A}_0)^{-1} \delta_t \quad (50)$$

where we want to find the mixing matrix  $(\mathbf{I} - \mathbf{A}_0)^{-1}$ . The general idea is that non-gaussianity allows us to use information from higher-order statistics to break the rotational equivalence of different SVARs. The logic of ICA is as follows. As demonstrated earlier, two different rotations will yield the same covariance matrix. When the innovations are gaussian, this renders the two rotated SVARs to be equivalent since gaussian variables are fully described by their first two moments. However, different rotations might induce different higher-order statistics in the non-gaussian case (Moneta et al., 2013). When the

structural innovations are non-gaussian and independent, we can thus distinguish between the following two cases:

$$\boldsymbol{\zeta}_t = (\mathbf{I} - \mathbf{A}_0)^{-1} \boldsymbol{\delta}_t \quad (51)$$

$$\boldsymbol{\zeta}_t = (\mathbf{I} - \mathbf{A}_0)^{-1} \mathbf{P}^\top \boldsymbol{\delta}_t \quad (52)$$

which were previously indistinguishable because they yield the same reduced-form innovation covariance. Under a further assumption that the structural innovations are independent across time, ICA proceeds by maximising the log-likelihood of the data:

$$\log L(\boldsymbol{\zeta}_1, \dots, \boldsymbol{\zeta}_T) = \sum_{t=1}^T \sum_{i=1}^K \log p_i\left(\frac{\mathbf{e}_i^\top (\mathbf{I} - \mathbf{A}_0) \boldsymbol{\zeta}_t}{\sigma_i}\right) - T \sum_{i=1}^K \log \sigma_i + T \log |\det(\mathbf{I} - \mathbf{A}_0)| \quad (53)$$

where  $p_i(\cdot)$  refers to the density of the  $i^{th}$  independent component (i.e. shock) that has been normalized to have unit variance. The above expression can be obtained via a standard change of variable formula and by exploiting the independence of the structural shocks.

Some further simplifications are usually made. Firstly, it is assumed that  $\mathbf{A}_0$  corresponds to an acyclic graph. Thus, we can always permute  $\mathbf{A}_0$  such that it is lower triangular. Since the absolute determinant of a matrix is invariant to permutation, and the determinant of a lower triangular matrix is simply the product of its diagonals, we can drop the last term in (53) because the permuted matrix  $\mathbf{I} - \mathbf{A}_0^{perm}$  has only ones on its diagonal (Hyvarinen et al., 2010). Secondly, the non-gaussian densities  $p_i$  are usually not known. However, it has been shown in the ICA literature that consistent estimation is still possible if we approximate it with

$$\log p_i(s) \approx -\sqrt{2}|s| + \text{const} \quad (54)$$

Note that the full likelihood can be written in terms of the data  $\mathbf{y}_t$  by using the substitution  $\boldsymbol{\zeta}_t = \mathbf{y}_t - \sum_{i=1}^p (\mathbf{I} - \mathbf{A}_0)^{-1} \mathbf{A}_i \mathbf{y}_{t-i}$  in (53). Under this setup, one can then maximise the likelihood to find the structural VAR parameters. Lastly, the independent shocks can only be recovered up to permutation, sign and scaling from ICA estimation. This issue is however automatically resolved when we have a acyclic  $\mathbf{A}_0$  (Moneta et al., 2013).

The ICA framework provides a powerful alternative for the estimation of SVARs. Unlike model search procedures, there are guarantees of consistent estimation when the structural innovations are non-gaussian. The cost is that we need to make stronger assumptions of independence rather than just uncorrelatedness. The ICA approach also shares many of the same limitations as model search procedures. Namely, it explicitly relies on an acyclic structure in  $\mathbf{A}_0$  and the shocks recovered also suffer from issues of interpretation. Note that when a acyclic structure cannot be assumed, heuristic rules are available to select a unique  $\mathbf{A}_0$  from within the equivalence class of contemporaneous matrices (Lanne et al., 2017). Nevertheless, these rules are mere heuristics and may not hold substantive meaning.

**Markov-Switching Heteroskedasticity.** Instead of assuming non-gaussianity, one could also identify SVARs by exploiting the possibility of heteroskedasticity in the data (Herwartz & Lütkepohl, 2014; Lanne et al., 2010; Lütkepohl & Woźniak, 2020). In this case, it is assumed that there is some joint process  $(\mathbf{y}_t, s_t)$  where  $\mathbf{y}_t$  is the observed data and  $s_t$  is a latent discrete-valued state. Specifically,  $s_t$  follows a markov process where

$$p(s_t \mid s_{1:t-1}) = p(s_t \mid s_{t-1}) \quad (55)$$

The key assumption for identification lies in the connection between  $s_t$  and  $\mathbf{y}_t$ . We assume that the reduced-form innovations exhibit state-dependent heteroskedasticity:

$$\boldsymbol{\zeta}_t \mid s_t = m \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_m) \quad (56)$$

Identification can then be obtained from a fact on the simultaneous diagonalizability of covariance matrices. Two covariance matrices  $\boldsymbol{\Sigma}_1$  and  $\boldsymbol{\Sigma}_2$  can always be written as  $\boldsymbol{\Sigma}_1 = \mathbf{W}\mathbf{W}^\top$  and  $\boldsymbol{\Sigma}_2 = \mathbf{W}\mathbf{D}\mathbf{W}^\top$ , where  $\mathbf{D}$  is a diagonal matrix with positive elements (Lanne et al., 2010). The columns in the matrix  $\mathbf{W}$  are unique up to sign changes and the decomposition is also unique up to a permutation of the variables. Note that the exact same  $\mathbf{W}$  appears in both decompositions. We can apply this decomposition to the conditional covariances of the reduced-form innovations  $\boldsymbol{\Sigma}_{\zeta|s_t=1}, \boldsymbol{\Sigma}_{\zeta|s_t=2}$  and set  $\mathbf{B}_0 = \mathbf{W}^{-1}$ .



Accordingly, the structural shocks defined by  $\mathbf{B}_0$  will satisfy the requirement of orthogonality since  $\text{Cov}(\boldsymbol{\delta}_t \mid s_t = m) = \text{Cov}(\mathbf{B}_0 \boldsymbol{\zeta}_t \mid s_t = m) = \mathbf{B}_0 \mathbf{W} \boldsymbol{\Lambda}_m \mathbf{W}^\top (\mathbf{B}_0)^\top = \boldsymbol{\Lambda}_m$ , where  $\boldsymbol{\Lambda}_1 = \mathbf{I}$  and  $\boldsymbol{\Lambda}_2 = \mathbf{D}$ . We thus have an unambiguous way of decomposing the reduced-form VAR into an SVAR. A similar result holds (with some qualifications) when there are more than 2 states. Suppose that the following decomposition

$$\boldsymbol{\Sigma}_i = \mathbf{W} \boldsymbol{\Lambda}_i \mathbf{W}^\top \quad \forall i \in \{1, \dots, M\} \quad (57)$$

holds and that for each  $k \neq j$ , there exists an  $i \in \{2, \dots, M\}$  such that  $\lambda_{ik} \neq \lambda_{ij}$ , where  $\lambda_{ij}$  is the  $j^{\text{th}}$  element of  $\boldsymbol{\Lambda}_i$ . Then, we have the result that the matrix  $\mathbf{W}$  is unique up to sign and permutation (Lanne et al., 2010). Note that such a decomposition is not always possible when there are more than 2 latent states, but statistical tests are available to check such an assumption against the data (Lanne et al., 2010; Lütkepohl & Woźniak, 2020). An SVAR model identified in this manner is also called a Markov-Switching SVAR (MS-SVAR). The model can be estimated by maximising the likelihood with a variant of the EM algorithm (Lanne et al., 2010).

The MS-SVAR model has mostly been used in econometrics. However, it may have certain use cases in psychology where the observed time-series may be driven by multiple states. For example, there is substantial evidence that rs-fMRI activity patterns differ depending on whether the person's eyes are open or closed (Agcaoglu et al., 2019; Tagliazucchi & Laufs, 2014). When participants are instructed to keep their eyes open, they may nevertheless end up dozing off mid-scan and reawaken only sometime later. One might be able to argue that a discretized version of alertness (e.g. awake, drowsy, asleep) may result in heteroskedasticity in the innovations, thereby allowing for identification of the SVAR. Similarly, a daily diary study among people with bipolar disorder may naturally result in heteroskedasticity due to participants switching between the manic and depressive phases (F. M. Bos et al., 2022).

Despite its seeming utility, the MS-SVAR framework suffers from important limitations. The most immediate problem is that the matrix  $\mathbf{B}_0$  is only identified up to a

permutation of the variables — it is unclear which shock maps onto which variable. This can be seen in (57) where rearrangement of the columns of  $\mathbf{W}$  coupled with a corresponding permutation of the diagonal of  $\mathbf{\Lambda}_i$  results in two different decompositions of the same covariance matrix. In short, we have orthogonal shocks that lack meaningful labels. Conventional MS-SVAR analyses in economics solve this permutation problem by inspecting the time-profiles of the estimated latent states. For example, in a SVAR analysis of oil prices, one might find that a given state has a high probability of occurrence in the late 1990s, which coincides with the Persian Gulf War. Such external information can then be used in labelling the nature of the shocks (i.e. oil supply shock). Alternatively, one could further impose zero restrictions to identify the shocks (Herwartz & Lütkepohl, 2014). Neither of these strategies are credible in psychology. In general, we rarely have information during particular windows of measurement beyond the measurements under study themselves, although the advent of passive sensing might ameliorate this. This makes it difficult to label the shocks if the MS-SVAR is applied to a psychology setting. This inability to resolve the permutation problem makes it difficult to conduct downstream analysis to relate the IRFs to person characteristics. One could nevertheless appeal to an acyclic structure in  $\mathbf{B}_0$ , but with the same caveats regarding acyclicity mentioned earlier.

It might also be relevant to point out that the MS-SVAR model is closely related to the ICA approach — conditional heteroskedasticity in the innovations imply non-gaussianity (Hyvarinen et al., 2010). Nevertheless, unlike ICA, the MS-SVAR identification scheme does not require independence (both contemporaneously and across time) and has testable implications when there are 3 or more markovian states. Thus, the MS-SVAR model has certain practical benefits and also allow for identification (up to permutation) under somewhat weaker conditions.

### ***Sign Restrictions***

The discussion thus far has mostly been focused on data-driven approaches to SVAR identification. While these can work well under stringent restrictions, there are often

serious issues with the interpretation of the resulting IRFs. The shocks found via data-driven techniques often are not automatically endowed with psychological meaning. This issue is further compounded by the possibility that shocks for the same variable have qualitatively different interpretations across different individuals. This complicates the scientific content of impulse response analysis in SVAR models.

In light of such issues, one alternative is to instead start off by specifying the nature of the shocks of interest at the very outset. This can be done by imposing sign restrictions on the impact multiplier matrix  $\mathbf{B}_0^{-1}$  (Kilian & Lütkepohl, 2017). Consider a motivating example based on an EMA study that investigated mental health during the COVID-19 pandemic (Fried et al., 2022). Three variables are of interest: time spent online, loneliness and time spent outdoors. Such a trivariate SVAR can be represented as:

$$\begin{bmatrix} (\text{Social Online})_t \\ (\text{Alone})_t \\ (\text{Outdoors})_t \end{bmatrix} = \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} + \begin{bmatrix} + & + & - \\ - & + & - \\ - & + & + \end{bmatrix} \begin{bmatrix} (\Delta \text{'Online Activity'})_t \\ (\Delta \text{'Workplace Disruption'})_t \\ (\Delta \text{'Time Outdoors'})_t \end{bmatrix} \quad (58)$$

The impact multiplier matrix here is specified up to sign. These explicit sign constraints reflect the nature of the shocks we are interested in studying. For example, we assume that a positive shock to time spent on social media online (e.g. received notification of a ‘like’ on your social media post) increases time spent online but decreases loneliness and time spent outdoors. Conversely, workplace disruption due to the coronavirus (e.g. workplace temporarily closed due to COVID) is assumed to increase online activity, loneliness and one’s need to seek social connection outdoors. Similarly, we can conceive that a positive shock to time spent outdoors (e.g. an unplanned trip to the grocery store) reduces online presence, decreases loneliness and increases time spent outdoors. Note that the sign patterns in each column are distinct: none of the columns are simple negations of another (Kilian & Lütkepohl, 2017). These sign restrictions reduce ambiguity by clarifying the nature of the shocks under study. That is, we have explicitly endowed the shocks with specific meaning. The same sign pattern can be imposed across all individuals, making a

post-hoc between-persons analysis a much more well-defined exercise. Obviously, one can always pick another set of sign restrictions based on alternative justifications. This is a feature rather than a flaw of this approach. A different choice of sign restrictions simply correspond to a different set of structural questions.

Sign restrictions can also be used to identify only a subset of shocks in a given VAR system. Consider the following extended example:

$$\begin{bmatrix} (\text{C19 Worry})_t \\ (\text{C19 Occupied})_t \\ (\text{Social Online})_t \\ (\text{Social Offline})_t \\ (\text{Alone})_t \\ (\text{Home})_t \\ (\text{Outdoors})_t \end{bmatrix} = \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} + \begin{bmatrix} + & - & * & * & * & * & * \\ + & - & * & * & * & * & * \\ + & + & * & * & * & * & * \\ - & - & * & * & * & * & * \\ + & - & * & * & * & * & * \\ * & + & * & * & * & * & * \\ - & - & * & * & * & * & * \end{bmatrix} \begin{bmatrix} (\Delta \text{Self-Isolation})_t \\ (\Delta \text{'Online Activity'})_t \\ * \\ * \\ * \\ * \\ * \end{bmatrix} \quad (59)$$

In the system above, seven variables were measured across time. Two items measured momentary concerns regarding COVID: responses to ‘C19 Worry’ reflect worry about the coronavirus and responses to ‘C19 Occupied’ reflect how occupied one was with regard to the pandemic. The variables ‘Social Online’ and ‘Social Offline’ refer to time spent on social media and offline social interactions respectively. The variable ‘Alone’ assessed one’s feelings of loneliness. The last two variables were related to time spent at home and outdoors. We are now interested in two shocks: the impact of self-isolation and a shock to time spent on social media online. All other shocks are labelled as ‘\*’ because they are not of interest. Accordingly, we have only specified sign restrictions for the two shocks, leaving all other values in the impact multiplier matrix to be unconstrained. Note that we left the impact of self-isolation on time spent at home to be unspecified since self-isolation can occur either at home or in a medical facility. This is possible because the pattern of restrictions present in the other rows is sufficient for us to distinguish between the two shocks even without knowing the sign of the impact of self-isolation on time spent at home.

The estimation and identification of these sign-restricted SVARs is based on a fairly simple idea. Since sign restrictions are relatively weak, it should be expected that point identification of the structural parameters is not generally possible (Kilian & Lütkepohl, 2017). Instead, we can aim for set identification — finding a set of structural parameters that are consistent with the stated sign restrictions. This can be done in two generic steps. Firstly, given the (estimated) reduced-form innovation covariance, we can apply the cholesky decomposition to obtain  $\Sigma_\zeta = \mathbf{L}\mathbf{L}^\top$ . Such a cholesky decomposition is clearly not theoretically meaningful, but it provides us with candidate orthogonal shocks via  $\zeta_t = \mathbf{L}\delta_t$ . Since the set of SVARs is related via rotation, having  $\zeta_t = \mathbf{L}\mathbf{P}\delta_t^*$  with  $\delta_t^* = \mathbf{P}^\top\delta_t$  yields another valid SVAR with orthogonal shocks. We can thus generate a large number of such SVARs by randomly generating rotation matrices and checking the signs of the resulting impact multiplier matrix  $\mathbf{L}\mathbf{P}$ . If the sign restrictions are violated, we discard the draw. In other words, we retain only the ones that satisfy the pattern of sign restrictions (Inoue & Kilian, 2013; Kilian & Lütkepohl, 2017). Of course, this approach only works if we have an efficient method of generating random orthogonal matrices. Fortunately, this can be done by exploiting a relation between the multivariate normal and orthogonal matrices. Let  $\mathbf{V}$  be a  $K \times K$  matrix where each element is drawn from an independent standard normal distribution. If we apply the QR decomposition such that  $\mathbf{V} = \mathbf{Q}\mathbf{R}$  with the diagonals of the upper triangular of  $\mathbf{R}$  constrained to be positive, then  $\mathbf{Q}$  is uniformly distributed over the space of  $K \times K$  orthogonal matrices (Rubio-Ramírez et al., 2010). Setting  $\mathbf{P} = \mathbf{Q}$  then provides us with a candidate impact multiplier matrix. This fact allows us to approximate the set of admissible SVARs with a large number of random rotations. This idea can be similarly applied when there are only a subset of shocks of interest. We simply check against the available sign restrictions for the relevant shocks, while making sure that the sign patterns of the columns for the remaining unspecified shocks can be distinguished from one another. For example,  $(+, +)$  and  $(-, -)$  are not distinguishable sign patterns, whereas  $(+, +)$  and  $(-, +)$  are.

Although we can obtain a set of SVAR models that are consistent with the sign restrictions, there remains the practical question of obtaining an appropriate summary of this set for downstream analysis. Two solutions to this problem are possible, both of which are based on Bayesian VARs (Inoue & Kilian, 2013; Kilian & Lütkepohl, 2017). Unlike estimation via OLS, the Bayesian approach treats the reduced-form parameters as random and specifies a prior over these parameters. Inference is then typically carried out via the posterior distribution. The conventional approach is to specify a normal-inverse Wishart prior on the parameters. Specifically, let  $\bar{\Phi} = [\Phi_1, \dots, \Phi_p]$ . Then the prior can be specified as:

$$\text{vec}(\bar{\Phi}) \mid \Sigma_\zeta \sim \mathcal{N}(\mathbf{0}, \mathbf{V} \otimes \Sigma_\zeta) \quad (60)$$

$$\Sigma_\zeta \sim \mathcal{IW}(\mathbf{S}_*, \nu) \quad (61)$$

where  $\mathbf{V}$ ,  $\mathbf{S}_*$  and  $\nu$  are the fixed parameters of the prior. This prior forms a conjugate pair with the gaussian likelihood, thus yielding a posterior distribution that is also normal-inverse Wishart and is available in closed-form (Kilian & Lütkepohl, 2017). The exact form of the posterior is omitted to avoid notational clutter. In the SVAR context, we also supplement these priors with a uniform prior over the set of orthogonal rotation matrices. Thus, a given draw from the posterior will yield some  $(\tilde{\Phi}, \tilde{\Sigma}_\zeta, \tilde{\mathbf{P}})$  and can be used to compute the structural IRF directly. Specifically, we can use  $\tilde{\Phi}$  to compute the Moving Average matrices and apply the cholesky decomposition to  $\tilde{\Sigma}_\zeta$ , followed by a rotation through  $\tilde{\mathbf{P}}$  to obtain  $(\tilde{\mathbf{B}}_0)^{-1}$ . If the resulting  $(\tilde{\mathbf{B}}_0)^{-1}$  satisfies the sign restrictions, the IRF can then be computed via (42). Repeating this for multiple draws from the posterior will yield a posterior of IRFs. If only a single summary statistic (e.g. time to equilibrium) for each IRF is required, then we can simply compute the desired statistic for each draw of the posterior and use a measure of central tendency for the resulting posterior over the statistic. Inference is more challenging when we are interested in the shape of the IRF. In such a case, one approach is to find the most likely IRF from the posterior. Fortunately, it turns out that an explicit formula for the posterior of the IRF (up to a

proportionality constant) is available, so we can use said formula to rank order the generated IRFs to find the modal response function (Inoue & Kilian, 2013).

**Limitations of Sign-Restricted SVARs.** Despite their potential utility, Sign-restricted SVARs suffer from several important limitations. Firstly, the posterior distributions of the IRFs can be very wide due to the weak nature of sign restrictions. This makes it difficult to make strong inferential statements. For example, most applications of Bayesian sign-restricted SVARs report 68% credible intervals instead of the conventional 95% intervals (Inoue & Kilian, 2013; Kilian & Lütkepohl, 2017). This is because the 95% credible intervals are usually too wide for meaningful statements to be made. Secondly, sign restrictions can be challenging to elicit for large VAR systems. It requires explicit consideration of the nature of the shocks and their relation to the variables under study and the specification of a large number of sign restrictions can be cumbersome. Moreover, it might be difficult to specify sign restrictions when a priori theoretical information is not available. This is often more of a problem in neuroscience than psychology where it might not be intuitively obvious whether a shock to a given neural region should be expected to increase or decrease activity in said area. Lastly, the use of non-conjugate priors or non-gaussian likelihoods makes it challenging to select the modal IRF from the posterior. While samples from the multivariate posterior are readily available, finding the multivariate mode is a non-trivial optimization problem. Other ad-hoc summaries from the marginal posterior of the IRFs at each horizon are possible, but they are problematic because the resulting metrics do not usually correspond to any admissible SVAR (Inoue & Kilian, 2013). This limits the use of sign-restricted SVARs to continuous data.

### Recommendations

The study of impulse responses in a VAR framework can produce important insights into the dynamics of human behavior. However, there is a certain level of nuance that needs to be taken into account when using impulse responses. These nuances are often glossed over in empirical practice. In this section, we provide some key recommendations

for the use of impulse response analysis in psychological practice.

Perhaps the most straightforward recommendation is to inspect the signs of the impact multiplier matrix prior to downstream between-persons analysis. This applies to all of the SVAR modeling approaches described so far. Obviously, the signs are consistent across people by fiat when the SVAR is identified via sign restrictions. This is not automatically the case when using data-driven approaches. A quick intermediate step is to simply check if there are consistencies in the signs of the individual impact multiplier matrices across people. If consistencies for a given shock can be found, then post-hoc analyses can proceed. Otherwise, caution should be warranted. Such a sanity check is often skipped over in practice.

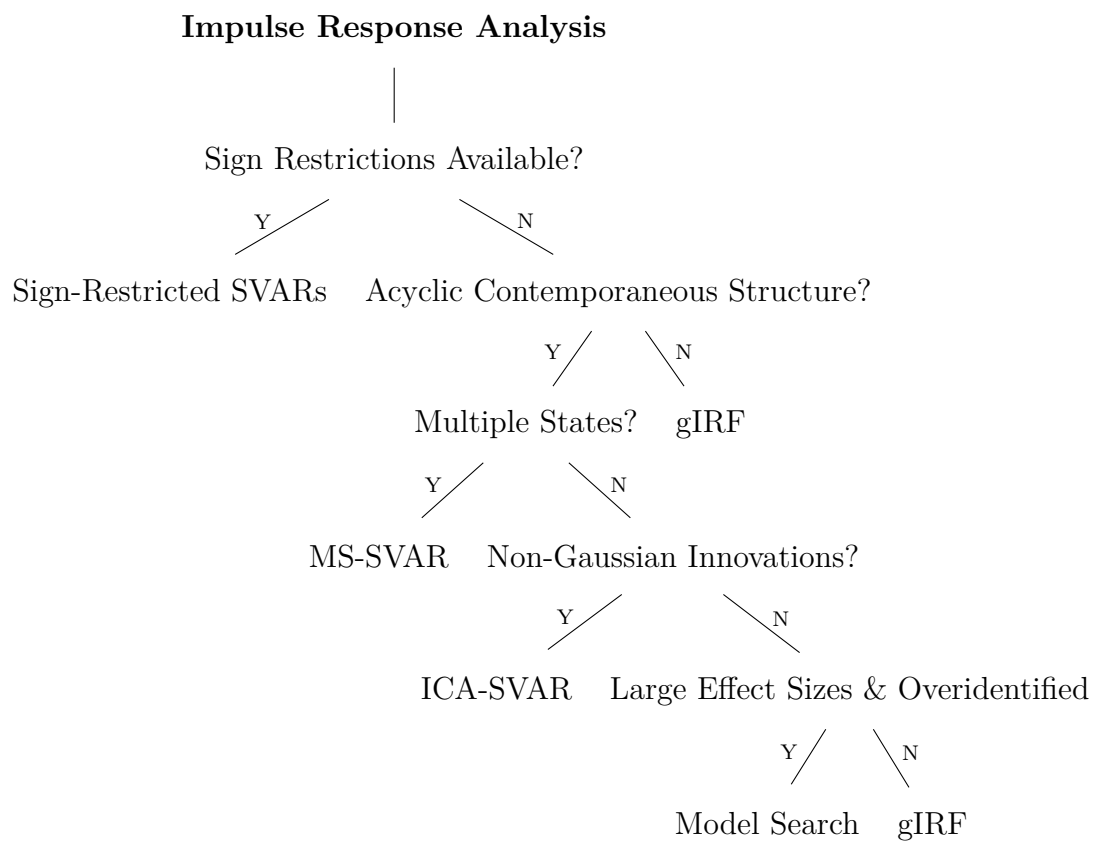
Another practical recommendation is to explicitly check the lag order of the VAR model. This is important because the choice of lag-order can substantially impact the IRF (Kilian & Lütkepohl, 2017). In psychology, a VAR(1) model is most commonly employed, with the main argument being that the sampling frequency used in many studies (e.g. daily diary of emotions) is too slow for any lagged effects beyond the first lag to be of interest (Krone et al., 2018). While this may be true in some areas of psychology, it is unlikely to be true in general. For example, there is evidence that a VAR(1) model is not necessarily appropriate for fMRI data, but SVAR analyses applied to fMRI data almost always assume a lag order of one (Weigard et al., 2023). Moreover, IRF analyses using a VAR(1) have been conducted even on psychological time-series data that was sampled every second (Yang et al., 2019). In this case, it is hard to believe that temporal effects do not persist beyond a single second. Similarly, modern EMA studies have allowed for the collection of time-series data with varying sampling frequencies. For example, EMA prompts may happen every hour in some studies but only occur every day in a different study. A VAR(1) model seems more realistic for the latter than for the former. A lag-order of one is an assumption to be checked (or justified), and should not be an automatic rule — using a wrong lag-order may result in distorted results in IRF analyses.



Finally, one should be acutely aware of the strengths and limitations underlying the different approaches to SVAR modelling. Our recommendations are summarized as a decision tree in Figure 1. The basic premise is that the assumptions of any SVAR modelling approach should be taken into account in the modelling process. For instance, assumptions such as acyclicity should be given due consideration given the process under study. If such an assumption is not satisfied, the uncritical application of data-driven approaches can produce misleading results. Even if acyclicity can be assumed, the nature of the problem may also guide the specific kind of data-driven search to be used. Conversely, one should consider if there is sufficient substantive knowledge available at hand to allow for the use of sign-restricted SVARs. When such information is available, a purely data-driven approach may in fact be less defensible given the stringent implicit assumptions required. It may also be the case that none of the assumptions for SVAR modelling are defensible. In such cases, we should abandon the idea of obtaining structural impulses and instead keep to a weaker kind of inference as captured by the gIRF. The key point is that we essentially need some a priori structural information, be they implicit or explicit, to proceed with a structural interpretation of a VAR.

### Conclusion

The VAR model is a workhorse in the study of psychological dynamics. To probe this model, impulse response analysis and the associated SVAR models have been developed to gain further insight into the model dynamics. In the psychological literature, SVARs were developed as a natural extension of the cross-sectional SEM; in econometrics, parallel developments resulted in alternative identification schemes to infer both structural and non-structural impulse responses. In either case, the study of impulse responses offer a unique window into the temporal evolution of psychological phenomena. That being said, uncritical use of impulse responses can be problematic. In reviewing the literature, we hope to improve how impulse responses are used in studying the temporal dynamics of the human mind.

**Figure 1**

*Decision tree for choice of SVAR modeling strategy.*

## References

- Agcaoglu, O., Wilson, T. W., Wang, Y.-P., Stephen, J., & Calhoun, V. D. (2019). Resting state connectivity differences in eyes open versus eyes closed conditions. *Human brain mapping, 40*(8), 2488–2498.
- Beltz, A. M., & Gates, K. M. (2017). Network mapping with GIMME. *Multivariate Behavioral Research, 52*(6), 789–804.
- Beltz, A. M., & Molenaar, P. C. M. (2016). Dealing with Multiple Solutions in Structural Vector Autoregressive Models. *Multivariate Behavioral Research, 51*(2-3), 357–373. <https://doi.org/10.1080/00273171.2016.1151333>
- Bos, F. M., Schreuder, M. J., George, S. V., Doornbos, B., Bruggeman, R., Van der Krieke, L., Haarman, B. C., Wichers, M., & Snippe, E. (2022). Anticipating manic and depressive transitions in patients with bipolar disorder using early warning signals. *International Journal of Bipolar Disorders, 10*(1), 12.
- Bos, F., Blaauw, F., Snippe, E., Van Der Krieke, L., De Jonge, P., & Wichers, M. (2018). Exploring the emotional dynamics of subclinically depressed individuals with and without anhedonia: An experience sampling study. *Journal of Affective Disorders, 228*, 186–193. <https://doi.org/10.1016/j.jad.2017.12.017>
- Demiralp, S., Hoover, K. D., & Perez, S. J. (2008). A Bootstrap Method for Identifying and Evaluating a Structural Vector Autoregression. *Oxford Bulletin of Economics and Statistics*. Retrieved September 29, 2025, from <https://onlinelibrary.wiley.com/doi/10.1111/j.1468-0084.2007.00496.x>
- Duffy, K. A., Fisher, Z. F., Arizmendi, C. A., Molenaar, P. C., Hopfinger, J., Cohen, J. R., Beltz, A. M., Lindquist, M. A., Hallquist, M. N., & Gates, K. M. (2021). Detecting Task-Dependent Functional Connectivity in Group Iterative Multiple Model Estimation with Person-Specific Hemodynamic Response Functions. *Brain Connectivity, 11*(6), 418–429. <https://doi.org/10.1089/brain.2020.0864>

- Fried, E. I., Papanikolaou, F., & Epskamp, S. (2022). Mental health and social contact during the covid-19 pandemic: An ecological momentary assessment study. *Clinical Psychological Science*, 10(2), 340–354.
- Gates, K. M., & Molenaar, P. C. (2012). Group search algorithm recovers effective connectivity maps for individuals in homogeneous and heterogeneous samples. *NeuroImage*, 63(1), 310–319. <https://doi.org/10.1016/j.neuroimage.2012.06.026>
- Gates, K. M., Molenaar, P. C., Hillary, F. G., Ram, N., & Rovine, M. J. (2010). Automatic search for fMRI connectivity mapping: An alternative to Granger causality testing using formal equivalences among SEM path modeling, VAR, and unified SEM. *NeuroImage*, 50(3), 1118–1125. <https://doi.org/10.1016/j.neuroimage.2009.12.117>
- Gunther, K. E., Edelman, A., Petrie, D., Kober, H., Gee, D. G., Joormann, J., & Gadassi-Polack, R. (2024). Dynamics between affect and social acceptance as a function of social anxiety: A person-specific network approach. *Emotion*.
- Hamaker, E. L., Dolan, C. V., & Molenaar, P. C. (2005). Statistical modeling of the individual: Rationale and application of multivariate stationary time series analysis. *Multivariate Behavioral Research*, 40(2), 207–233.
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.
- Herwartz, H., & Lütkepohl, H. (2014). Structural vector autoregressions with Markov switching: Combining conventional with statistical identification of shocks. *Journal of Econometrics*, 183(1), 104–116. <https://doi.org/10.1016/j.jeconom.2014.06.012>
- Hyvarinen, A., Hyvarinen, A., Zhang, K., Zhang, K., Shimizu, S., Hoyer, P. O., & Hoyer, P. (2010). Estimation of a Structural Vector Autoregression Model Using Non-Gaussianity. *Journal of Machine Learning Research*.
- Inoue, A., & Kilian, L. (2013). Inference on impulse response functions in structural VAR models. *Journal of Econometrics*, 177(1), 1–13. <https://doi.org/10.1016/j.jeconom.2013.02.009>

- Jongerling, J., Laurenceau, J.-P., & Hamaker, E. L. (2015). A multilevel AR (1) model: Allowing for inter-individual differences in trait-scores, inertia, and innovation variance. *Multivariate Behavioral Research*, 50(3), 334–349.
- Kilian, L., & Lütkepohl, H. (2017). *Structural vector autoregressive analysis*. Cambridge University Press.
- Kim, J., Zhu, W., Chang, L., Bentler, P. M., & Ernst, T. (2007). Unified structural equation modeling approach for the analysis of multisubject, multivariate functional MRI data. *Human Brain Mapping*, 28(2), 85–93.  
<https://doi.org/10.1002/hbm.20259>
- Koop, G., Pesaran, M., & Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1), 119–147.  
[https://doi.org/10.1016/0304-4076\(95\)01753-4](https://doi.org/10.1016/0304-4076(95)01753-4)
- Krone, T., Albers, C. J., Kuppens, P., & Timmerman, M. E. (2018). A multivariate statistical model for emotion dynamics. *Emotion*, 18(5), 739–754.  
<https://doi.org/10.1037/emo0000384>
- Lanne, M., Lütkepohl, H., & Maciejowska, K. (2010). Structural vector autoregressions with Markov switching. *Journal of Economic Dynamics and Control*, 34(2), 121–131. <https://doi.org/10.1016/j.jedc.2009.08.002>
- Lanne, M., Meitz, M., & Saikkonen, P. (2017). Identification and estimation of non-Gaussian structural vector autoregressions. *Journal of Econometrics*, 196(2), 288–304. <https://doi.org/10.1016/j.jeconom.2016.06.002>
- Liu, S. (2017). Person-specific versus multilevel autoregressive models: Accuracy in parameter estimates at the population and individual levels. *British Journal of Mathematical and Statistical Psychology*, 70(3), 480–498.
- Lütkepohl, H., & Woźniak, T. (2020). Bayesian inference for structural vector autoregressions identified by Markov-switching heteroskedasticity. *Journal of*

*Economic Dynamics and Control*, 113, 103862.

<https://doi.org/10.1016/j.jedc.2020.103862>

Lydon-Staley, D. M., Cornblath, E. J., Blevins, A. S., & Bassett, D. S. (2021). Modeling brain, symptom, and behavior in the winds of change. *Neuropsychopharmacology*, 46(1), 20–32. <https://doi.org/10.1038/s41386-020-00805-6>

Lydon-Staley, D. M., Leventhal, A. M., Piper, M. E., Schnoll, R. A., & Bassett, D. S. (2021). Temporal networks of tobacco withdrawal symptoms during smoking cessation treatment. *Journal of Abnormal Psychology*, 130(1), 89–101. <https://doi.org/10.1037/abn0000650>

Molenaar, P. C. M. (2017). Equivalent Dynamic Models. *Multivariate Behavioral Research*, 52(2), 242–258. <https://doi.org/10.1080/00273171.2016.1277681>

Moneta, A., Entner, D., Hoyer, P. O., & Coad, A. (2013). Causal Inference by Independent Component Analysis: Theory and Applications. *Oxford Bulletin of Economics and Statistics*. Retrieved September 29, 2025, from <https://onlinelibrary.wiley.com/doi/10.1111/j.1468-0084.2012.00710.x>

Nestler, S., & Humberg, S. (2021). GIMME’s ability to recover group-level path coefficients and individual-level path coefficients. *Methodology*, 17(1), 58–91.

Pearl, J. (2009). *Causality*. Cambridge university press.

Pesaran, H., & Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1), 17–29. [https://doi.org/10.1016/S0165-1765\(97\)00214-0](https://doi.org/10.1016/S0165-1765(97)00214-0)

Rubio-Ramírez, J. F., Waggoner, D. F., & Zha, T. (2010). Structural Vector Autoregressions: Theory of Identification and Algorithms for Inference. *Review of Economic Studies*, 77(2), 665–696. <https://doi.org/10.1111/j.1467-937X.2009.00578.x>

- Seizer, L., Fuchs, D., Bliem, H. R., & Schubert, C. (2023). Emotional states predict cellular immune system activity under conditions of life as it is lived: A multivariate time-series analysis approach. *Plos One*, 18(11), e0290032.
- Seong, B., Morshed, A. M., & Ahn, S. K. (2006). Additional sources of bias in half-life estimation. *Computational Statistics & Data Analysis*, 51(3), 2056–2064.
- Sims, C. A. (1980). Macroeconomics and Reality. *Econometrica*, 48(1), 1–48.  
<https://doi.org/10.2307/1912017>
- Swanson, N. R., & Granger, C. W. J. (1997). Impulse Response Functions Based on a Causal Approach to Residual Orthogonalization in Vector Autoregressions. *Journal of the American Statistical Association*, 92(437), 357–367.  
<https://doi.org/10.1080/01621459.1997.10473634>
- Tagliazucchi, E., & Laufs, H. (2014). Decoding wakefulness levels from typical fMRI resting-state data reveals reliable drifts between wakefulness and sleep. *Neuron*, 82(3), 695–708.
- Weigard, A., Lane, S., Gates, K., & Beltz, A. (2023). The influence of autoregressive relation strength and search strategy on directionality recovery in group iterative multiple model estimation. *Psychological Methods*, 28(2), 379–400.  
<https://doi.org/10.1037/met0000460>
- Wright, A. G. C., Hopwood, C. J., & Simms, L. J. (2015). Daily Interpersonal and Affective Dynamics in Personality Disorder. *Journal of Personality Disorders*, 29(4), 503–525. Retrieved September 29, 2025, from <https://www.proquest.com/docview/1697863940/abstract/F9AFDE7C4B784B2BPQ/1>
- Yang, X., Ram, N., Gest, S. D., Lydon-Staley, D. M., Conroy, D. E., Pincus, A. L., & Molenaar, P. C. M. (2018). Socioemotional Dynamics of Emotion Regulation and Depressive Symptoms: A Person-Specific Network Approach. *Complexity*.  
<https://doi.org/10.1155/2018/5094179>

Yang, X., Ram, N., Lougheed, J. P., Molenaar, P. C. M., & Hollenstein, T. (2019).

Adolescents' emotion system dynamics: Network-based analysis of physiological and emotional experience. *Developmental Psychology*, 55(9), 1982–1993.

<https://doi.org/10.1037/dev0000690>